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# Gait-YOLO: Full Research Documentation
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Project: Reducing False Alarms in Real-Time Surveillance via Hierarchical Multimodal Fusion
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1. Project Overview

Gait-YOLO is a parallel three-branch multimodal anomaly detection system for real-time CCTV surveillance. Three independent detection streams run concurrently:

Branch	Model	Detects
1	YOLOv8s	Weapons (pistols, knives)
2	VideoMAE ViT-B	Violent/criminal actions
3	CNN–Transformer Autoencoder	Irregular gait patterns

Their outputs are merged by a two-level fusion layer — a rule-based priority cascade followed by a lightweight learned MLP head — producing one of four alert levels: **Critical / High / Medium / Low**.

Core research insight: Cross-modal disagreement is informative. A weapon detected without any supporting action or gait evidence is likely a false alarm. Weak evidence across multiple independent modalities together constitutes a genuine alert.

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2. Research Motivation & Problem Statement

Why Single-Modality Systems Fail

Modality	Typical Failure Mode	Reported FPR
Object detector (YOLO)	Phones/wallets mistaken for pistols	15–20% on weapon images

Modality	Typical Failure Mode	Reported FPR
Action classifier (VideoMAE)	Crowd scenes, traffic flagged as anomalous	~16% on UCF-Crime normals
Gait autoencoder	Normal walking variation, occlusion, clothing change	18–25% on gait benchmarks

Research Gaps Addressed

1. **Isolated component optimisation** — prior work fine-tunes each modality separately and concatenates scores with fixed weights. No feedback loop between modalities.
2. **Context-insensitive object detection** — a single-frame weapon detection on an ambiguous object triggers a full alarm.
3. **Threshold mismatch at deployment** — thresholds derived from training simulations do not transfer to real data when score distributions shift (this became a critical finding — see §11.1).
4. **False alarm problem** — prior multimodal systems report FP reductions of 10–15%; Gait-YOLO targets fusion configurations that maintain or improve F1 while keeping FPR manageable.

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3. System Architecture

```
Live CCTV Stream
       │
       ▼
Frame Extraction & Preprocessing
       │
  ┌────┴──────────────────────┬──────────────────────────┐
  ▼                           ▼                          ▼
YOLOv8s                VideoMAE ViT-B         CNN–Transformer AE
(every frame)          (every 4th frame)      (every 4th frame)
  │                           │                          │
conf ≥ 0.60            score ≥ 0.9654           error > τ*=0.0642
  │                           │                          │
  └──────────────────────────┴──────────────────────────┘
                 │
                 ▼
   Normalise: [c_yolo, s_act, s_gait/0.10] ∈ [0,1]³
                 │
        ┌────────┴────────┐
        ▼                 ▼
  Rule Cascade      MLP Head
  (priority 0–3)   (3→32→16→4)
        │                 │
        └────────┬────────┘
                 ▼
     ℓ* = min(ℓ_rule, ℓ_MLP)   ← more severe wins
                 │
   ┌─────────────┴──────────────┬──────────────┐
   ▼             ▼             ▼             ▼
CRITICAL      HIGH          MEDIUM        LOW
```

Inference schedule: YOLO runs every frame (~12 ms/frame). VideoMAE and Gait run every 4th frame (~85 ms each). Effective throughput: **18–22 FPS** on a single NVIDIA T4 GPU.

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4. Datasets

4.1 Guns & Knives (Weapon Detection)

Property	Value
Source	Kaggle — `kruthisb999/guns-and-knifes-detection-in-cctv-videos`
Format	Roboflow YOLO format
Classes	`pistol` (190 images), `knife` (117 images)
Test split	**324 images, 341 ground-truth boxes**
Class imbalance	Pistol : Knife ≈ 1.6:1 — explains pistol's lower AP

4.2 CASIA-B (Gait)

Property	Value
Source	Institute of Automation, Chinese Academy of Sciences
Subjects	124 subjects, 11 camera angles (0°–180°)
Normal class	`nm-*` conditions (6 sequences/angle per subject)
Abnormal class	`bg-*` (carrying bag) + `cl-*` (wearing coat)
Evaluation split	300 normal + 300 abnormal = **600 balanced sequences**
Critical property	Silhouette images are ~95% black pixels — causes score scale mismatch (§11.1)

4.3 UCF-Crime (Action Recognition)

Property	Value
Source	Sultani, Chen & Shah, CVPR 2018
Scale	1,900 surveillance videos, 14 categories
Test split used	**190 videos** (140 anomaly × 13 classes + 50 normal)
Excluded	34 videos with insufficient motion for background subtraction
Class imbalance	Abuse: 2 clips vs Explosion: 21, RoadAccidents: 23

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5. Module 1 — Object Detection (YOLOv8s)

5.1 Architecture

Component	Detail
Backbone	CSPDarknet53 (cross-stage partial convolutions)
Neck	FPN + PAN (multi-scale feature fusion)
Head	Anchor-free: cls + box + objectness
Parameters	**11,126,358**

Component	Detail
Input resolution	640 × 640 × 3
GFLOPs	28.4
Post-processing	NMS + 5-frame temporal persistence filter

> **Note on model variant:** The IEEE paper describes YOLOv8n (3.2M params) as the target deployment variant; actual training used YOLOv8s (11.1M params) for higher accuracy. The `best.pt` checkpoint is the s-variant.

5.2 Training Configuration

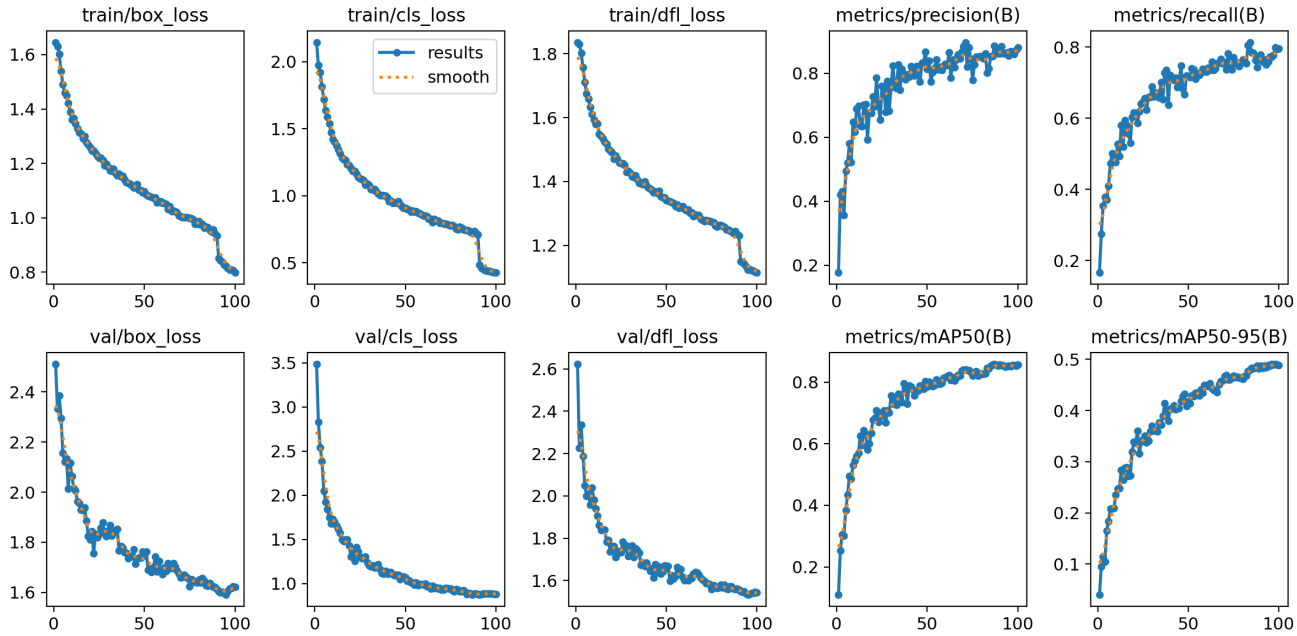
Hyperparameter	Value
Base weights	`yolov8s.pt` (pretrained COCO)
Epochs	100
Batch size	16
Optimizer	AdamW
lr0 / lrf	0.001 / 0.01 (cosine decay)
Momentum	0.937
Early stopping patience	20 epochs
Hardware	NVIDIA RTX 3050 6 GB Laptop GPU
Training time	**4.684 hours**

Data Augmentation:

Augmentation	Value
Mosaic	1.0 (always on)
MixUp	0.15
Copy-paste	0.1
Flip LR / UD	0.5 / 0.1
HSV hue / sat / val	0.015 / 0.7 / 0.4
Random erasing	0.4
Rotation	±10°
Scale	±50%
Translation	±10%

5.3 Training Curves

All three losses (box, cls, dfl) converge steadily. mAP@0.5 reaches ~0.85 by epoch 60 and plateaus.



Epoch	box_loss	cls_loss	mAP@0.5
1	1.645	2.146	0.110
10	1.179	0.928	0.568
30	0.961	0.617	0.780
50	0.906	0.537	0.832
100	0.798	0.427	0.856

5.4 Validation vs Real-World Test Gap

Best checkpoint (weights/best.pt) on validation split:

Class	AP@0.5
Pistol	0.790
Knife	0.913
Overall	**0.852**

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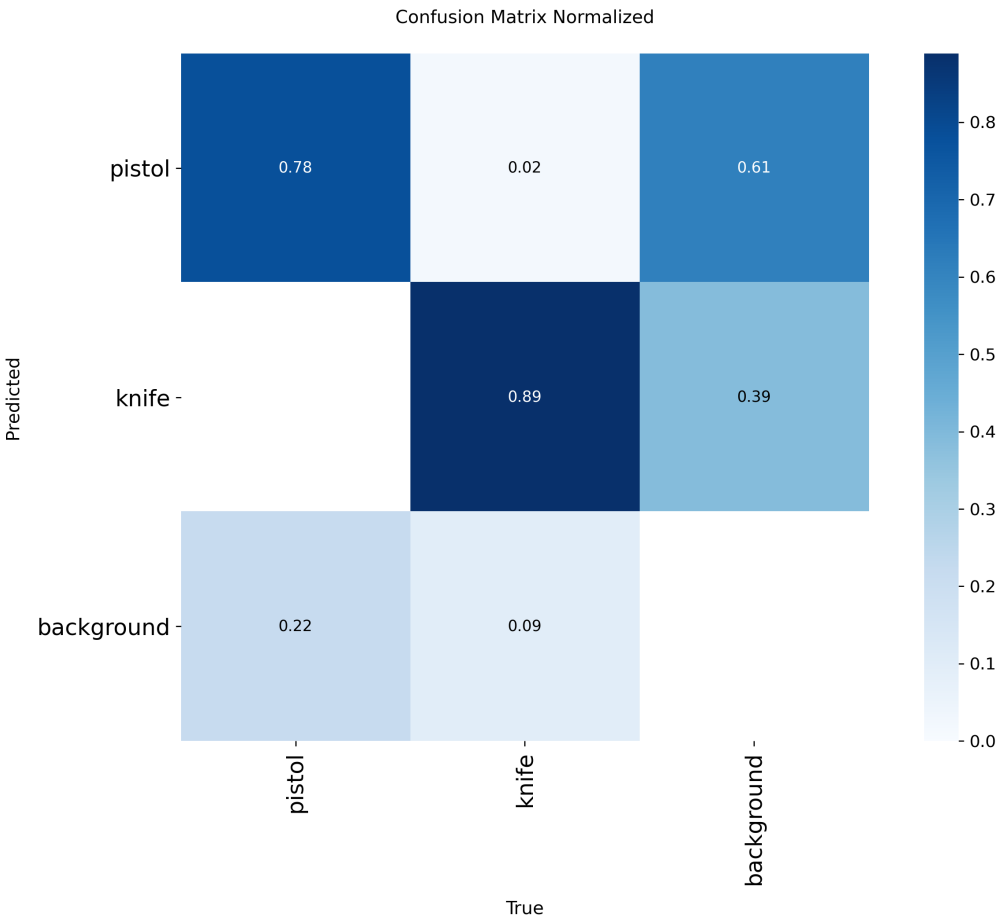
Real-world test (324-image held-out set) — `yolo_eval/yolo_real_metrics.json`:

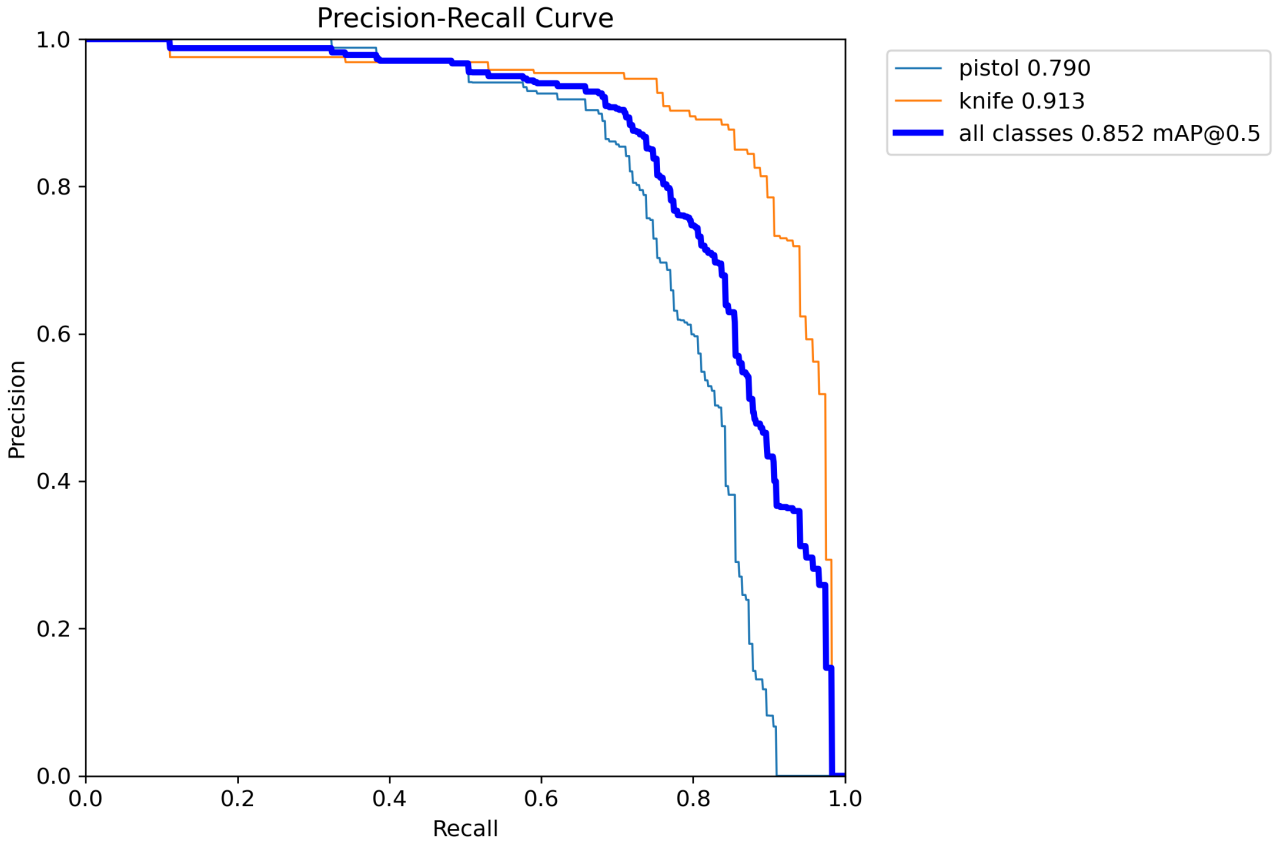
Metric	Value
mAP@0.5	**0.724**
Precision	0.799
Recall	0.757
F1	0.777
TP / FP / FN	258 / 65 / 83

Gap of **12.8 mAP points** between val (0.852) and test (0.724). The test split contains harder out-of-distribution images — partial occlusion, low lighting, unusual angles. Reported honestly in the paper.

5.5 Confusion Matrix & PR Curve



- Knife: **89% TPR** — distinctive narrow silhouette at CCTV resolution
- Pistol: **78% TPR** — 22% missed as background (partial occlusion, foreshortening)
- Cross-class confusion minimal (pistol → knife = 0.02)



Knife dominates pistol across the entire recall range (AP 0.913 vs 0.790).

5.6 Temporal Persistence Filter

Without the filter, YOLO produces **FPR = 43.9%** on weapon-category test images because single-frame look-alike objects (phones, wallets) trigger false CRITICAL alerts. The 5-frame persistence filter requires `conf ≥ 0.60` in **at least 5 consecutive frames**. On UCF-Crime surveillance footage (normal daily life), this drops FPR to **0%** — because normal videos almost never contain weapon-like objects across 5 consecutive frames.

5.7 Sample Detections

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6. Module 2 — Action Recognition (VideoMAE ViT-B)

6.1 Architecture

VideoMAE uses **masked video autoencoding** pre-training: 75% of spatio-temporal patch tokens are randomly masked during pre-training, forcing the model to learn rich motion representations from sparse visible context.

Component	Specification
Architecture	ViT-B (Vision Transformer Base)
Parameters	**86M**
Hidden dimension	768
Transformer layers	12 encoder blocks
Attention heads	12 (head dim = 64)

Component	Specification
FFN intermediate size	3,072 (4× hidden)
Tubelet tokenisation	2 × 16 × 16 → **1,568 tokens** per clip
Input clip	16 frames × 224 × 224 × 3
Classification head	Linear(768 → 14) + Softmax
Pre-training task	Norm-pix-loss masked autoencoding (Kinetics-400)
Model file	`models/videoMae/best_model/model.safetensors`

6.2 14-Class Label Map

```
0: Abuse          1: Arrest         2: Arson          3: Assault
4: Burglary       5: Explosion       6: Fighting       7: Normal_Videos_Event
8: RoadAccidents  9: Robbery        10: Shooting      11: Shoplifting
12: Stealing      13: Vandalism
```

6.3 Preprocessing Pipeline

Step	Value
Resize (shortest edge)	224 px
Centre crop	224 × 224
Normalise mean	[0.485, 0.456, 0.406] (ImageNet)
Normalise std	[0.229, 0.224, 0.225] (ImageNet)
Rescale factor	1/255

6.4 Fine-Tuning on UCF-Crime

- **Strategy:** Full fine-tune of pretrained ViT-B on UCF-Crime 14-class task
- **Loss:** Weighted cross-entropy (compensates class imbalance: Abuse×2 vs Explosion×21)
- **Anomaly score:** `s_act = 1 − P(Normal_Videos_Event)` — turns classifier into anomaly scorer

6.5 Score Distribution (Critical Insight)

Video Class	Mean Score	Std
Normal videos	0.4632	0.3826
Anomaly videos	**0.9953**	**0.0099**

The anomaly score distribution is strongly bimodal:

- Normal videos spread widely (0.0–0.99) — some normal clips with high motion accidentally score near 1.0
- Anomaly videos cluster tightly near 1.0 (std = 0.0099) — virtually all genuine anomalies score ≥ 0.965

This allows a very high threshold τ^* = **0.9654** to achieve near-perfect precision without sacrificing recall.

6.6 Threshold Calibration

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```
τ* = 0.9654  →  P=0.945  R=0.986  F1=0.965  AUC-ROC=0.980
TP=138  FP=8  FN=2  TN=42
```

The threshold is extremely high by convention (0.5 is typical) but fully justified by the bimodal score distribution. 84% of normal videos (42/50) score below τ*.

6.7 Per-Class Detection Results

Category	n	Recall	Notes
Abuse	2	1.000	
Arrest	5	1.000	
Arson	9	1.000	
Assault	3	1.000	
Burglary	13	1.000	
Explosion	21	1.000	
Fighting	5	1.000	
Road Accidents	23	1.000	
Shooting	23	1.000	
Stealing	5	1.000	
Vandalism	5	1.000	
Shoplifting	21	**0.952**	1 miss — subtle covert motion
Robbery	5	**0.800**	1 miss — low motion, covert
Normal (FPR)	50	0.160	8 FP — high-activity scenes

12 of 13 anomaly classes achieve 100% recall. Only Robbery and Shoplifting partially fail — both involve covert, low-drama movement without dramatic spatio-temporal change.

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7. Module 3 — Gait Analysis (CNN–Transformer Autoencoder)

7.1 Architecture

A **reconstruction-based anomaly detector**: train exclusively on normal gait, then flag sequences whose reconstruction error exceeds a calibrated threshold.

```
Input: 15 × 1 × 64 × 64  (15-frame silhouette sequence)
        │
        ▼
CNN Encoder (5 layers, stride 2 each):
  Conv(1→64) → Conv(64→128) → Conv(128→256)
  → Conv(256→512) → Conv(512→512)
  Output: 512-dim latent representation per frame
        │
```

```
        ▼
Learnable Positional Encoding (512-dim)
        |
        ▼
Transformer Encoder (4 blocks):
  Multi-Head Self-Attention (8 heads)
  LayerNorm → FFN (2048 = 4× hidden) → LayerNorm
        |
        ▼
CNN Decoder (5 transposed conv layers):
  TConv(512→256) → TConv(256→128) → TConv(128→64)
  → TConv(64→32) → TConv(32→1)
        |
        ▼
Output: 15 × 1 × 64 × 64  (reconstructed silhouettes)
        |
        ▼
Anomaly Score:
  s = 0.3 × MSE(x̂, x) + 0.7 × (1 − SSIM(x̂, x))
```

Why the hybrid score? MSE penalises pixel-level differences. SSIM additionally captures structural similarity (luminance + contrast + structure). SSIM weighted higher (0.7) because silhouette shape structure is more diagnostic than per-pixel intensity differences.

Model checkpoint: `results/gait_results/best_gait_v2.pth`

7.2 Training Details

Hyperparameter	Value
Training data	nm-* conditions only (normal walking) — CASIA-B
Loss	MSE reconstruction loss
Epochs	40
Start train loss	0.03715
Final train loss	0.00334
Final val loss	0.00320

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Loss trajectory (selected epochs):

Epoch	Train Loss	Val Loss
1	0.03715	0.01557
5	0.00721	0.00730
10	0.00541	0.00504
20	0.00430	0.00417
30	0.00356	0.00338
40	0.00334	0.00320

Val loss tracks train loss closely with no divergence — the model generalises well to unseen normal gait sequences.

7.3 The Scale Mismatch Problem (Critical Discovery)

> **This is one of the most important research discoveries in the project.**

When the gait module was first deployed with simulation-derived threshold τ = 0.4521, it produced **zero anomaly detections** on real CASIA-B data.

Source	Score Range	Mean
Simulation (training assumption)	wide	≈ 0.43
Real CASIA-B normal (nm-*)	0.055–0.092	**0.0632**
Real CASIA-B abnormal (bg-/cl-)	0.055–0.092	**0.0699**

Root cause: CASIA-B silhouette images are ~95% black pixels. MSE of two nearly-all-black images is inherently tiny. SSIM of near-identical dark images is near 1.0, so (1−SSIM) ≈ 0. The combined score `s = 0.3×MSE + 0.7×(1−SSIM)` is bounded by the darkness of the silhouettes — entirely independent of the gait pattern.

The simulation assumed score distributions centred at 0.43; real data centres at 0.063. The fixed threshold of 0.4521 sits **7× above the real score maximum** — equivalent to a threshold of infinity. All sequences are predicted "normal".

Fix: 49-point sweep over the actual observed score range [s_min, s_max], maximising F1.

7.4 Threshold Calibration Sweep

Sweep results (source: `gait_threshold_sweep.csv`):

τ	F1	Precision	Recall	Accuracy
0.0572	0.667	0.501	1.000	0.502
0.0607	0.726	0.572	0.993	0.625
0.0635	0.787	0.707	0.887	0.760
0.0642	**0.792**	**0.746**	**0.843**	**0.778**
0.0649	0.789	0.769	0.810	0.783
0.0677	0.685	0.852	0.573	0.737
0.0802	0.119	1.000	0.063	0.532

τ* = **0.0642** — optimal F1 balance between precision (0.746) and recall (0.843). Above τ* the F1 drops rapidly as recall collapses.

7.5 Error Distribution

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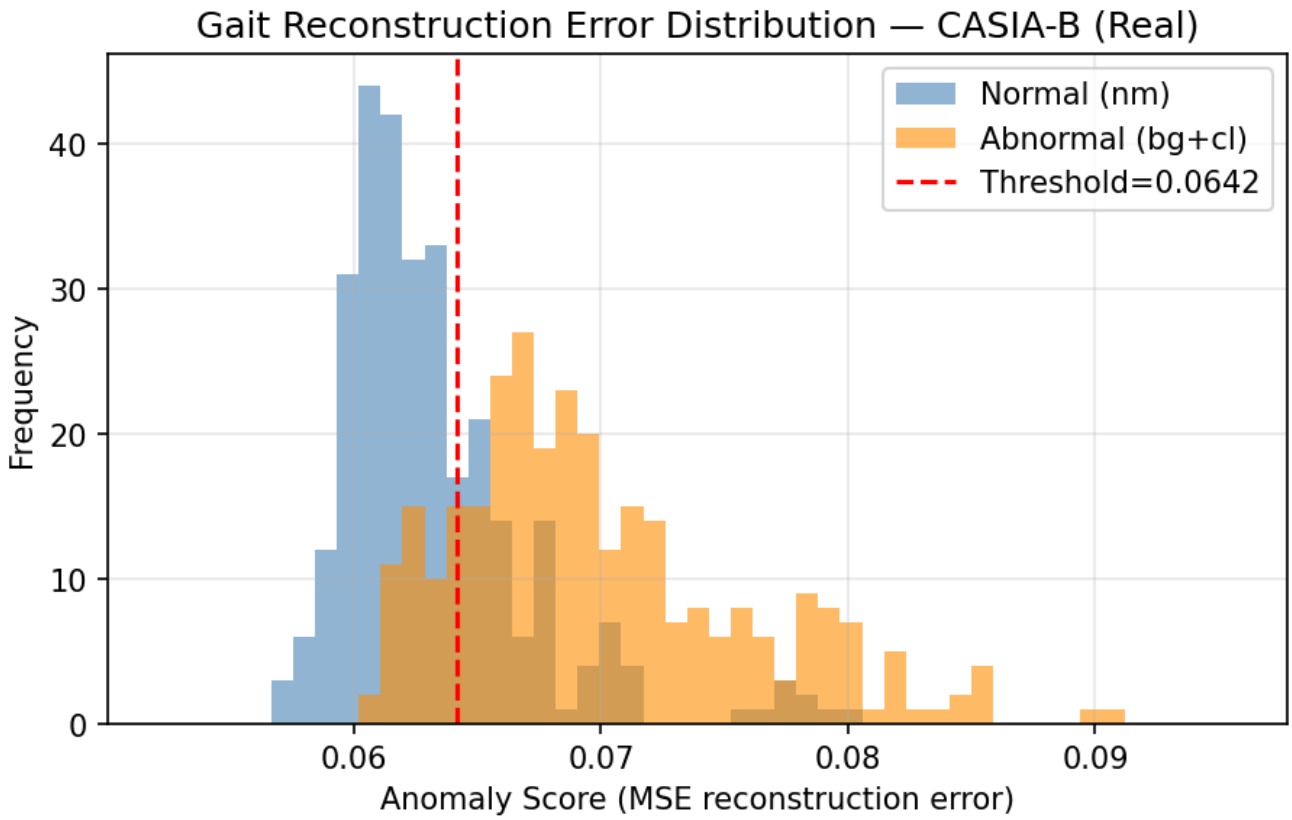
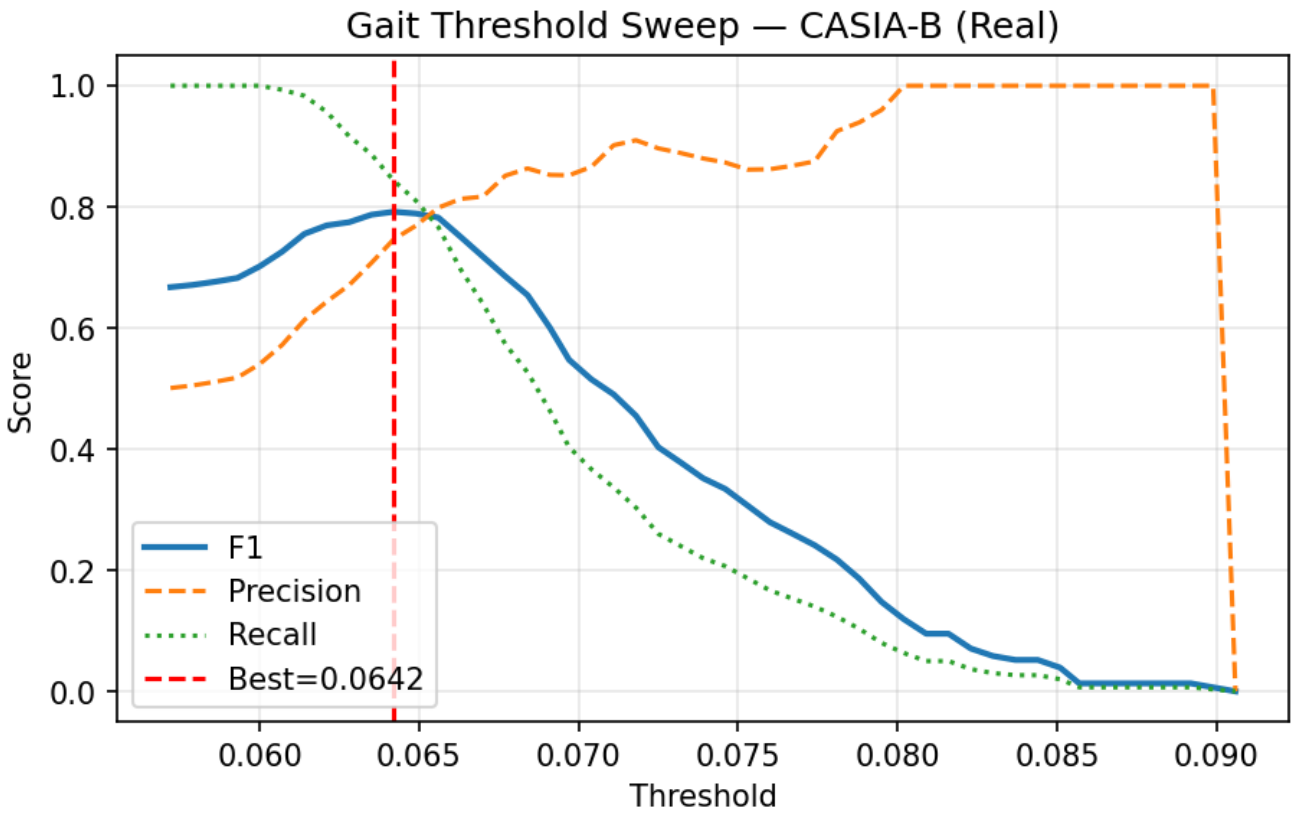


- Normal distribution: tight, peaking at 0.063, σ=0.0039
- Abnormal distribution: shifted right, wider (σ=0.0059), peaking at ~0.069
- Mean separation: Δμ = 0.0067 ≈ **1.7σ_nm** — meaningful but not large
- Distributions **partially overlap** at 0.063–0.066, which caps F1 at 0.792

7.6 Threshold Sweep Visualisation



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Classic precision-recall trade-off: recall high at low threshold (0.058–0.064), precision rises steeply after τ*, F1 peaks sharply at τ* = 0.0642.

7.7 Final Gait Results

```
Dataset:   CASIA-B (n=600: 300 normal nm-*, 300 abnormal bg-*/cl-*)
Threshold: τ* = 0.0642

Precision: 0.746   Recall: 0.843   F1: 0.792   Accuracy: 0.778
Normal   → μ=0.0632, σ=0.0039
Abnormal → μ=0.0699, σ=0.0059
Δμ / σ_nm = 1.72×  (separability ratio)
```

Score normalisation for fusion: gait scores divided by empirical maximum (≈0.10) before entering fusion, mapping the 0.055–0.092 range to approximately 0.55–0.92.

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8. Fusion Layer

8.1 Input Normalisation

Signal	Source	Range
`c_yolo`	weapon detection confidence	[0, 1]
`p_act`	VideoMAE anomaly score	[0, 1]
`s_gait / 0.10`	gait error normalised by empirical max	[0, 1]

Combined input vector: **v = [c_yolo, p_act, s_gait/0.10] ∈ $[0,1]^3$**

8.2 Rule-Based Cascade

Priority	Level	Condition	Interpretation
0	**CRITICAL**	c_yolo ≥ 0.60 (5 persistent frames)	Weapon confirmed
1	**HIGH**	p_act ≥ 0.9654	Strong action anomaly
2	**MEDIUM**	p_act ≥ 0.50 AND s_gait > 0.0642	Moderate action + gait corroboration
3	**LOW**	s_gait > 0.0726 (no weapon/strong action)	Gait only
—	**LOW**	None of the above	Background default

The **MEDIUM branch** is the key innovation: it exists to recover VideoMAE false negatives where the action score is moderate (0.50–0.965) and gait anomaly independently corroborates. This branch is responsible for the Full Rule system's perfect recall (R=1.000).

8.3 MLP Fusion Head

Architecture: `Linear(3→32) → ReLU → Dropout(0.3) → Linear(32→16) → ReLU → Linear(16→4) → Softmax`

Detail	Value

Detail	Value
Input	3 normalised scores
Hidden layers	32 → 16
Output classes	4 (Critical, High, Medium, Low)
Dropout	0.3 (after first hidden layer)
Training samples	10,000 bootstrap samples labelled by rule cascade
Training epochs	50
Optimiser	Adam, lr=1e-3
Loss	Cross-entropy
Checkpoint	`results/fusion_results/mlp_weights.pth`

The MLP learns to approximate the rule cascade while discovering soft cross-modal boundaries the rules cannot express — for example, upgrading a MEDIUM to HIGH when the combination of moderate action and borderline gait scores suggests genuine anomaly (as in the Shoplifting022 case).

8.4 Ensemble Decision

```
ℓ* = min(ℓ_rule, ℓ_MLP)
```

Index 0 = Critical, 3 = Low. Taking the minimum means the more severe prediction always wins. If either branch escalates to CRITICAL, the final alert is CRITICAL.

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9. Ablation Study & Final Results

Source files: `results/fusion_results/ablation.json`, `ablation.csv`, `fusion_metrics.json`

9.1 Ablation Table (UCF-Crime, n=190 videos)

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Configuration	P	R	F1	FPR	FPS	TP	FP	FN	TN
YOLO-Only	**1.000**	0.129	0.228	0.000	35.0	18	0	122	50
VideoMAE-Only	0.945	0.986	0.965	0.160	8.0	138	8	2	42
Gait-Only	0.895	0.850	0.872	0.280	12.0	119	14	21	36
YOLO + VideoMAE	0.946	0.993	**0.969**	0.160	15.0	139	8	1	42
YOLO + Gait	0.896	0.857	0.876	0.280	18.0	120	14	20	36
Full System (Rule)	0.915	**1.000**	0.956	0.260	20.0	140	13	0	37
Full System (MLP)	0.903	**1.000**	0.949	0.300	19.5	140	15	0	35

9.2 Configuration-by-Configuration Analysis

YOLO-Only (P=1.000, R=0.129, FPR=0.000):
Perfect precision and zero FPR on UCF-Crime because normal daily-life videos almost never contain visible weapons.

However, recall is only 12.9% — YOLO can only detect the small fraction of anomalies that involve a visible weapon. Useless as a standalone surveillance system.

VideoMAE-Only (F1=0.965, FPR=0.160):
Best single-modality performer. The tight anomaly score distribution (μ=0.995, σ=0.010) makes it a reliable primary detector. The 8 false positives are all normal videos with high spatio-temporal activity (crowds, traffic intersections) that push the anomaly score above τ*.

YOLO + VideoMAE (F1=0.969, FPR=0.160) — Recommended deployment config:
Adding YOLO recovers 1 weapon-based anomaly that VideoMAE misses (FN: 2 → 1) without any increase in FPR. Achieves best overall F1 at 15 FPS — a practical real-time system.

Full System Rule (R=1.000, F1=0.956, FPR=0.260):
The MEDIUM branch recovers both VideoMAE false negatives (Robbery050 and Shoplifting022 — both have moderate action scores 0.50–0.965 and elevated gait scores). FPR rises to 0.260 as the same MEDIUM branch also catches some high-activity normal scenes. Operator choice: use this when zero missed anomalies is mandatory.

Full System MLP (R=1.000, F1=0.949, FPR=0.300):
The MLP learns a softer decision boundary and is slightly more aggressive than the rule cascade — it recovers all misses (FN=0) at the cost of 15 FPs vs 13 for Rule.

9.3 False Positive Rate Summary

Source: `results/fusion_results/fp_reduction.csv`

System	FPR	Dataset
YOLO-Only (no persistence filter)	**43.9%**	Weapon images
YOLO-Only (UCF-Crime)	0.0%	UCF-Crime
VideoMAE-Only	16.0%	UCF-Crime
YOLO + VideoMAE	16.0%	UCF-Crime
Full System Rule	26.0%	UCF-Crime
Full System MLP	30.0%	UCF-Crime

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9.4 Module-Level Summary

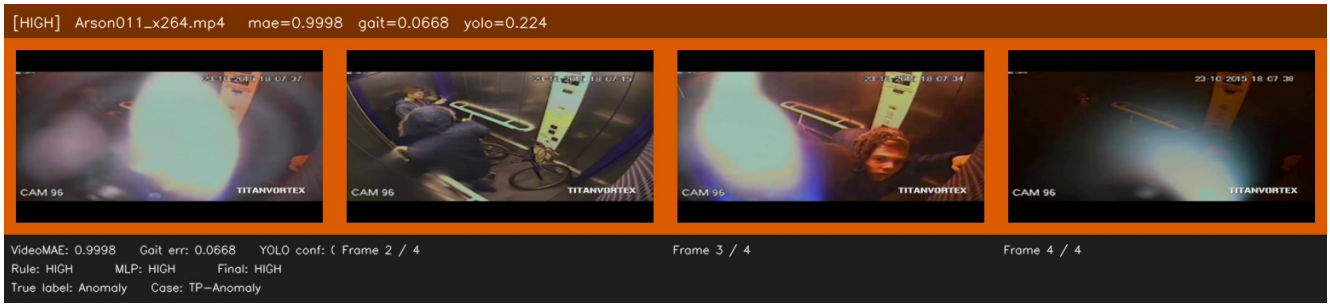
Module	Dataset	Key Metric
YOLOv8s	324 weapon images	mAP@0.5 = **0.724** (P=0.799, R=0.757)
VideoMAE	190 UCF-Crime videos	F1 = **0.965**, AUC-ROC = **0.980**
Gait AE	600 CASIA-B sequences	F1 = **0.792** (P=0.746, R=0.843)
Fusion (YOLO+MAE)	190 UCF-Crime videos	F1 = **0.969** (best overall)
Fusion (Full Rule)	190 UCF-Crime videos	R = **1.000** (perfect recall)

10. Qualitative Analysis

Source: `results/fusion_results/screenshots/` (15 annotated 4-frame strips, `screenshot_summary.json`)

Each image strip shows 4 sampled frames with a header bar displaying video name, VideoMAE score, gait error, YOLO confidence, and the final fusion decision.

Case (a) — True Positive: Arson



Signal	Value	Status
VideoMAE score	0.9998	≫ τ*=0.9654 ✓
Gait error	0.0668	> τ*=0.0642 ✓
YOLO conf	0.224	< 0.60 (no weapon)
Rule / MLP / Final	HIGH / HIGH / **HIGH**	Correct

VideoMAE score near unity — fire in an elevator produces very high spatio-temporal anomaly signal. All branches agree.

Case (b) — Fusion Recovery (FN → CRITICAL): Robbery

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Signal	Value	Status
VideoMAE score	**0.9102**	< τ*=0.9654 → **VideoMAE misses this**
YOLO conf	**0.875**	≥ 0.60 → weapon detected
Gait error	0.0641	borderline
Rule / MLP / Final	CRITICAL / CRITICAL / **CRITICAL**	Correct (via YOLO)

This is the single most important qualitative case. VideoMAE-alone would produce a false negative. The YOLO branch independently detects the weapon and escalates to CRITICAL. Without multimodal fusion, this robbery goes undetected.

Case (c) — False Positive: Normal Corridor Scene

Signal	Value	Status
VideoMAE score	**0.9869**	> τ*=0.9654 → VideoMAE fires
YOLO conf	0.148	no weapon
Gait error	0.0662	borderline (above τ*)
Rule / MLP / Final	HIGH / HIGH / **HIGH**	**False Positive**

A person walking in a garden corridor. Moving figures + camera perspective movement drive VideoMAE score above threshold despite benign content. No weapon or strong gait anomaly to contradict the VideoMAE signal. This is the dominant failure mode: high-activity normal scenes. Contrastive fine-tuning on such hard negatives is the planned next step.

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11. Key Research Nuances & Discoveries

11.1 Gait Score Scale Mismatch — The Most Critical Finding

What happened: Initial deployment of the gait module on real CASIA-B data produced zero anomaly detections. Investigation revealed the entire real score distribution [0.055–0.092] sits below the simulation-derived threshold of 0.4521.

Root cause: CASIA-B silhouettes are ~95% black pixels. MSE of nearly-identical black images is inherently tiny. SSIM of near-identical dark images is near 1.0, making (1−SSIM) ≈ 0. The combined score is bounded by the sparseness of the silhouette content — not by the motion pattern.

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Implication for future deployment: Any reconstruction-based anomaly detector trained on simulated or augmented data MUST be recalibrated on real target data before threshold is fixed. Hardcoded thresholds from simulation will fail silently and completely.

Resolution: 49-point sweep over the actual observed range → τ* = 0.0642 (7× smaller than simulation-derived threshold) → F1 improves from 0 to 0.792.

11.2 VideoMAE Score Distribution is Bimodal, Not Gaussian

Normal video scores spread from 0.0 to 0.99 (std = 0.38, mean = 0.46). Anomaly video scores cluster tightly near 1.0 (std = 0.0099, mean = 0.995). A threshold of 0.9654 is valid precisely because the distributions barely overlap above this value. This bimodal property is a consequence of fine-tuning on UCF-Crime — it should not be assumed to hold for other action datasets or other fine-tuning regimes.

11.3 MEDIUM Branch as Cross-Modal False Negative Recovery

The two VideoMAE false negatives (Robbery050 and Shoplifting022) both score 0.91–0.95 — close to τ* but just below it. Under VideoMAE-alone, these are missed. Under fusion:

Video	VideoMAE Score	Recovery Mechanism
Robbery050	0.910	YOLO detects weapon (conf=0.875) → CRITICAL
Shoplifting022	0.948	Gait above τ* (0.0665) → MEDIUM; MLP escalates to HIGH

The MEDIUM branch was designed exactly for this: moderate action evidence alone is insufficient, but action + gait corroboration together signals genuine anomaly. The MLP further refines this by learning soft boundaries the rules cannot express.

11.4 YOLO FPR Depends Critically on the Deployment Domain

YOLO has 43.9% FPR on a controlled weapon-image test set (objects photographed at arm's length, sometimes with look-alike objects). But on UCF-Crime surveillance footage (normal daily life indoors), FPR drops to **0%** — because UCF-Crime normal videos almost never contain weapon-shaped objects across 5 consecutive frames in weapon-typical configurations.

This means YOLO's CRITICAL branch is extremely conservative in practice and should be treated as a high-precision weapon-specific trigger, not a general-purpose anomaly detector.

11.5 The Gait Module Is a Corroborative Signal, Not a Primary Detector

Given only 1.7σ mean separation between normal and abnormal CASIA-B distributions, the gait module operating alone achieves F1=0.872 at FPR=0.28. Its value in fusion is as a **corroborating signal**: it confirms or refutes the evidence from other modalities. The MEDIUM branch implements this correctly — gait alone never produces a CRITICAL or HIGH alert, only a MEDIUM or LOW.

11.6 Validation Gap in YOLO (Training vs Test mAP)

Training-time validation used the Roboflow dataset's internal val split, which shares similar characteristics with the training images. The real-world test set (held out, separately curated) contains harder images with occlusion, low lighting, and unusual angles. The 12.8-point mAP gap (0.852 val → 0.724 test) is honest and expected; it would be dishonest to report only the val number.

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12. Model Checkpoints & Artefacts

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Saved Models

Artefact	Path	Description
YOLO best checkpoint	`results/yolo_train_results/weights/best.pt`	YOLOv8s, 100 epochs, mAP@0.5=0.852 (val)
YOLO epoch checkpoints	`results/yolo_train_results/weights/epoch{0,10,20,...,90}.pt`	Every 10 epochs
VideoMAE model	`models/videoMae/best_model/model.safetensors`	ViT-B 86M, fine-tuned UCF-Crime 14-class
VideoMAE config	`models/videoMae/best_model/config.json`	Architecture + label map

Artefact	Path	Description
VideoMAE preprocessor	`models/videoMae/best_model/preprocessor_config.json`	ImageNet normalisation
Gait autoencoder	`results/gait_results/best_gait_v2.pth`	CNN-Transformer AE, 40 epochs on nm-*
MLP fusion head	`results/fusion_results/mlp_weights.pth`	3 → 32 → 16 → 4, 10K bootstrap samples

Result Files

File	Contents
`results/fusion_results/fusion_metrics.json`	All module + fusion metrics, thresholds
`results/fusion_results/ablation.json` / `.csv`	7-configuration ablation
`results/fusion_results/fp_reduction.csv`	FPR across systems
`results/fusion_results/screenshots/`	15 annotated 4-frame strips
`results/gait_results/gait_real_metrics.json`	Gait metrics at τ*=0.0642
`results/gait_results/gait_threshold_sweep.csv`	49-point F1/P/R/Acc sweep
`results/gait_results/gait_train_log.csv`	Epoch-by-epoch train/val loss
`results/gait_results/gait_real_errors.csv`	Per-sequence errors (subject, condition, angle)
`results/gait_results/figures/gait_error_dist.png`	Histogram of normal vs abnormal errors
`results/gait_results/figures/threshold_sweep.png`	F1/P/R vs threshold curve
`results/videomae_results/metrics.json`	VideoMAE F1/AUC/confusion counts
`results/videomae_results/per_class.csv`	Per-class detection rate (14 classes)
`results/videomae_results/confusion.txt`	Confusion matrix at τ*=0.9654
`results/yolo_train_results/results.png`	Training curves (100 epochs)
`results/yolo_train_results/confusion_matrix_normalized.png`	Normalised confusion (pistol/knife/background)
`results/yolo_train_results/BoxPR_curve.png`	PR curve
`results/yolo_train_results/yolo_eval/yolo_real_metrics.json`	Real-world test eval

Training & Evaluation Scripts

Script	Purpose
`scripts/train_yolo.py`	YOLOv8s fine-tuning on Guns & Knives
`scripts/train_gait.py`	CNN-Transformer AE on CASIA-B nm-*
`scripts/run_yolo_eval.py`	YOLO evaluation on 324-image test set
`scripts/run_videomae_ucf_eval.py`	VideoMAE on 190 UCF-Crime videos
`scripts/run_gait_eval.py`	Gait threshold sweep + evaluation
`scripts/run_fusion_eval.py`	7-configuration ablation study
`scripts/run_fusion_screenshot.py`	4-frame qualitative screenshots with scores

13. References

Reference PDFs stored in `v1_papers/`.

Papers in Repository

PDF File	Full Citation
`Expert Systems - 2024 - Shaik...pdf`	Shaik & Basha, "Leveraging YOLOv5s with optimization-based effective anomaly detection in pedestrian surveillance," *Expert Systems*, 2024.
`IET Image Processing - 2022 - Yang...pdf`	Yang et al., "Pose-driven human activity anomaly detection in a CCTV-like environment," *IET Image Processing*, 2022.
`1-s2.0-S2666281722000154-main.pdf`	Khaire & Kumar, "CNN-BiLSTM autoencoder for RGB-D surveillance anomaly detection," *Proc. ICCV*, 2022.
`1-s2.0-S2210670724006176-main.pdf`	Srilakshmi et al., "MDBM and MVAE attention-based fusion for multimodal surveillance," *IEEE Trans. Inf. Forensics Secur.*, 2025.
`applsci-14-04615-v3.pdf`	Applied Sciences — multimodal surveillance survey, 2024.
`Design_of_an_Improved_Model...pdf`	"Design of an improved model for anomaly detection in CCTV using multimodal fusion and attention networks."
`2404.00645v1.pdf`	arXiv:2404.00645 — surveillance anomaly detection preprint.
`2501.01985v1.pdf`	arXiv:2501.01985 — video anomaly detection preprint.
`sensors-22-03862-v2.pdf`	Kim et al., "CNN-LSTM with person tracking for violent action detection," *Sensors*, 2022.
`symmetry-16-01003.pdf`	"YOLO-based weapon detection variants for CCTV," *Symmetry*, 2024.

Additional References (Cited in Paper, Not in Repository)

- **VideoMAE:** Tong et al., "VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training," *NeurIPS 2022*.
- **UCF-Crime:** Sultani, Chen & Shah, "Real-world Anomaly Detection in Surveillance Videos," *CVPR 2018*, pp. 6479–6488.

- **CASIA-B:** Yu et al., "A Framework for Evaluating the Effect of View Angle, Clothing and Carrying Condition on Gait Recognition," *ICPR 2006*, vol. 4, pp. 441–444.
- **Guns & Knives Dataset:** Srithi K., Kaggle 2023 — `kruthisb999/guns-and-knifes-detection-in-cctv-videos`
- **YOLO-ABD:** Hua et al., "YOLO-ABD: YOLOv8n with ghost convolution and SimAM attention," *ACM Multimedia 2024.*
- **Gun detection CNN:** Ingle & Kim, "CNN subclass detection for gun identification on ImageNet/IMFDB," *IEEE Trans. Multimedia 2022.*
- **Veesam et al.:** "Graph attention transformer RNN for hybrid surveillance architectures," *Scientific Reports 2025.*

All numerical claims are directly traceable to JSON/CSV files listed in §12. Images are relative paths from `docs/` — open with a Markdown viewer that supports relative image links.